Classification of Groups of Order pq

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of classification of groups of order pq, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

MA-03 / Modern Algebra / Report v1

I. TARGET AND IMPORT CONTRACT

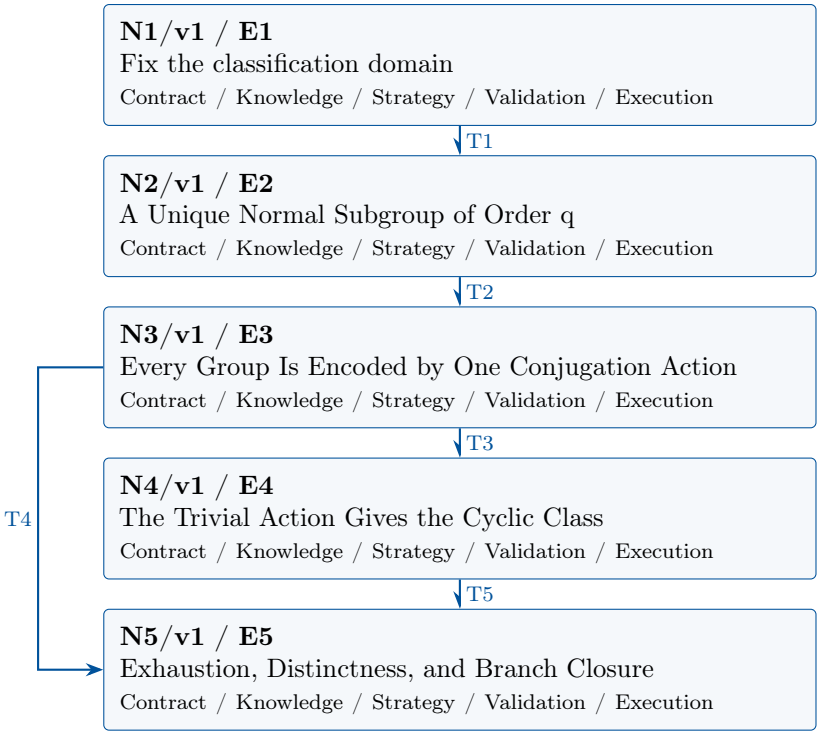
Theorem I.1. *Let $p < q$ be primes. If p does not divide $q - 1$, every group of order pq is isomorphic to C_{pq}. If p divides $q - 1$, there are exactly two isomorphism classes: C_{pq} and one nonabelian group $C_q \rtimes C_p$.*

A complete classification must prove that every input group has one of the stated structures, construct each listed structure, and prove that apparently different nontrivial constructions are isomorphic. Sylow congruences alone do not close these obligations.

A. Foundations and permitted imports

Sylow's existence/counting theorems; Lagrange's theorem; finite product-set counting; the theorem that the multiplicative group of the prime field F_q is cyclic of order $q - 1$. The last result is the supporting primitive-root module, not an unstated assumption. Prime-order groups are cyclic because a nonidentity element generates a subgroup whose nontrivial order divides that prime.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The two possible regimes are $p \nmid (q-1)$ and $p \mid (q-1)$ for $|G| = pq$, $p < q$ prime.

Receiving activity. N2: A Unique Normal Subgroup of Order q

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1; Sylow counting and subgroup product cardinalities.

T2: N2/E2/v1 to N3. $G = QP$ with $Q \cong C_q$ normal, $P \cong C_p$, and $Q \cap P = 1$.

Receiving activity. N3: Every Group Is Encoded by One Conjugation Action

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2 and cyclicity of prime-order groups.

T3: N3/E3/v1 to N4. Every group has normal forms $x^i y^j$ with $yxy^{-1} = x^r$ and $r^p \equiv 1 \pmod{q}$.

Receiving activity. N4: The Trivial Action Gives the Cyclic Class

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and cyclicity of $\mathbb{F}_q^\times$.

T4: N3/E3/v1 to N5. Every group has normal forms $x^i y^j$ with $yxy^{-1} = x^r$ and $r^p \equiv 1 \pmod{q}$.

Receiving activity. N5: Exhaustion, Distinctness, and Branch Closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and N4/E4, including the construction and equivalence certificates.

T5: N4/E4/v1 to N5. The trivial action gives C_{pq}; a nontrivial class exists exactly when $p \mid (q-1)$, and all nontrivial actions yield one isomorphism class.

Receiving activity. N5: Exhaustion, Distinctness, and Branch Closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and N4/E4, including the construction and equivalence certificates.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Fix the classification domain

Object and role. Fix the classification domain. Establish the following local result: The two possible regimes are $p \nmid (q-1)$ and $p \mid (q-1)$ for $|G| = pq$, $p < q$ prime.

Inputs. Distinct primes, finite groups, and the stated classification target.

C — Contract

Own E1/v1: The two possible regimes are $p \nmid (q-1)$ and $p \mid (q-1)$ for $|G| = pq$, $p < q$ prime.

K — Knowledge

Primes, subgroup orders, Sylow counts

S — Strategy

Fix classification obligations

V — Validation

Require $p < q$

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Fix the ordered primes and the order of the group.

N1.2 Require existence and exhaustion, not just permitted Sylow counts.

N1.3 Keep the cyclic class in both regimes.

Export E1/v1. The two possible regimes are $p \nmid (q-1)$ and $p \mid (q-1)$ for $|G| = pq$, $p < q$ prime.

Acceptance gate. Require $p < q$ Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Require $p < q$ Recheck dependent nodes N2, N3, N4, N5.

B. N2/v1: A Unique Normal Subgroup of Order q

Object and role. A Unique Normal Subgroup of Order q. Establish the following local result: $G = QP$ with $Q \cong C_q$ normal, $P \cong C_p$, and $Q \cap P = 1$.

Inputs. N1/E1; Sylow counting and subgroup product cardinalities.

C — Contract

Own E2/v1: $G = QP$ with $Q \cong C_q$ normal, $P \cong C_p$, and $Q \cap P = 1$.

K — Knowledge

Sylow q-count theorem

S — Strategy

Eliminate the larger candidate

V — Validation

Use count divides p

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Use $n_q \mid p$ and $n_q \equiv 1 \pmod{q}$ to force $n_q = 1$.

N2.2 Choose a Sylow p-subgroup and use coprime orders for trivial intersection.

N2.3 Count $|QP| = pq$ and conclude $QP = G$.

Export E2/v1. $G = QP$ with $Q \cong C_q$ normal, $P \cong C_p$, and $Q \cap P = 1$.

Acceptance gate. Use count divides p Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Use count divides p Recheck dependent nodes N3, N4, N5.

C. N3/v1: Every Group Is Encoded by One Conjugation Action

Object and role. Every Group Is Encoded by One Conjugation Action. Establish the following local result: Every group has normal forms $x^i y^j$ with $yxy^{-1} = x^r$ and $r^p \equiv 1 \pmod{q}$.

Inputs. N2/E2 and cyclicity of prime-order groups.

C **Contract**

Own E3/v1: Every group has normal forms $x^i y^j$ with $yxy^{-1} = x^r$ and $r^p \equiv 1 \pmod{q}$.

K **Knowledge**

Prime-order cyclic groups

S **Strategy**

Translate conjugation into exponent r

V **Validation**

Check trivial intersection and full product

E **Execution**

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Choose generators of Q and P.

N3.2 Encode conjugation as a nonzero residue r modulo q.

N3.3 Derive $(i, j)(k, l) = (i + r^j k, j + l)$ and the constraint $r^p = 1$.

Export E3/v1. Every group has normal forms $x^i y^j$ with $yxy^{-1} = x^r$ and $r^p \equiv 1 \pmod{q}$.

Acceptance gate. Check trivial intersection and full product Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Check trivial intersection and full product Recheck dependent nodes N4, N5.

D. N4/v1: The Trivial Action Gives the Cyclic Class

Object and role. The Trivial Action Gives the Cyclic Class. Establish the following local result: The trivial action gives C_{pq}; a nontrivial class exists exactly when $p \mid (q-1)$, and all nontrivial actions yield one isomorphism class.

Inputs. N3/E3 and cyclicity of $\mathbb{F}_q^\times$.

C — Contract

Own E4/v1: The trivial action gives C_{pq}; a nontrivial class exists exactly when $p \mid (q-1)$, and all nontrivial actions yield one isomorphism class.

K — Knowledge

Cyclic Fq multiplicative group

S — Strategy

Classify homomorphisms $Cp \to \mathrm{Aut}(Cq)$

V — Validation

Separate permission, existence, and equivalence

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 For $r = 1$, prove that commuting generators yield an element of order pq.

N4.2 For an order-p residue, verify associativity, identity, inverses, and noncommutativity of the pair multiplication.

N4.3 If $r' = r^k$ with $1 \le k < p$, use $(i, j) \mapsto (i, kj)$ to identify the two constructed groups.

Export E4/v1. The trivial action gives C_{pq}; a nontrivial class exists exactly when $p \mid (q-1)$, and all nontrivial actions yield one isomorphism class.

Acceptance gate. Separate permission, existence, and equivalence Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Separate permission, existence, and equivalence Recheck dependent nodes N5.

E. N5/v1: Exhaustion, Distinctness, and Branch Closure

Object and role. Exhaustion, Distinctness, and Branch Closure. Establish the following local result: There is one class if $p \nmid (q-1)$ and exactly two if $p \mid (q-1)$.

Inputs. N3/E3 and N4/E4, including the construction and equivalence certificates.

C — Contract

Own E5/v1: There is one class if $p \nmid (q-1)$ and exactly two if $p \mid (q-1)$.

K — Knowledge

All action and decomposition evidence

S — Strategy

Join exhaustive branches

V — Validation

Keep cyclic case in both regimes

E — Execution

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N5.1 Every original group is encoded by one admissible action.

N5.2 The action classification supplies all possibilities and no duplicate nontrivial class.

N5.3 Distinguish the cyclic and nonabelian classes by commutativity.

Export E5/v1. There is one class if $p \nmid (q-1)$ and exactly two if $p \mid (q-1)$.

Acceptance gate. Keep cyclic case in both regimes Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N5: Keep cyclic case in both regimes Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Fix the classification domain

Let $p < q$ be distinct primes and let G have order pq. The classification must establish existence, exhaustion, and uniqueness of every isomorphism class, separately in the two divisibility regimes.

B. N2 — A Unique Normal Subgroup of Order q

Let n_q be the number of Sylow q-subgroups. Sylow gives $n_q \mid p$ and $n_q \equiv 1 \pmod{q}$. Thus n_q is 1 or p. Since $1 < p < q$, p cannot be congruent to 1 modulo q. Hence $n_q = 1$, and its unique subgroup Q is normal.

Choose a Sylow p-subgroup P. Both Q and P are cyclic, of orders q and p respectively. Their intersection is trivial because its order divides both distinct primes. The product set QP has pq elements, hence equals G.

C. N3 — Every Group Is Encoded by One Conjugation Execution

Choose generators x of Q and y of P. Every g in G has a unique expression $x^i y^j$, with i modulo q and j modulo p. Conjugation by y preserves Q, so

$$yxy^{-1} = x^r$$

for nonzero r modulo q. Because $y^p = 1$, repeated conjugation gives $r^p \equiv 1 \pmod{q}$.

Multiplying normal forms gives

$$(x^i y^j)(x^k y^l) = x^{i+r^j k} y^{j+l}.$$

Thus the group is completely determined by the action parameter r, and every group in the problem has such a parameter. Its order in F_q multiplicative is either 1 or p.

D. N4 — The Trivial Action Gives the Cyclic Class

If $r = 1$, x and y commute. The element xy has order pq: if $(xy)^k = 1$, then $x^k = y^{(-k)}$ lies in $Q \cap P$, so both q and p divide k. Conversely $(xy)^{(pq)} = 1$. Therefore G is cyclic of order pq.

This class exists for every pair of primes, including when p divides $q - 1$.

1. Construct and Verify the Nontrivial Class

A nontrivial r with $r^p = 1$ has order p. Such r exists exactly when p divides $q - 1$, because F_q multiplicative is cyclic of that order.

For any such r, define a product on pairs $(i, j) \in \mathbb{Z}/q\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ by

$$(i, j)(k, l) = (i + r^j k, j + l).$$

The condition $r^p = 1$ makes the expression independent of the representative of j. Associativity holds because both ways of multiplying three pairs have first component $i + r^j k + r^{j+l} m$ and second component j+l+n. The identity is (0,0), and the inverse of (i,j) is $(-r^{-j} i, -j)$. Thus this is a group with pq elements.

For $x = (1, 0)$, $y = (0, 1)$, we have $yx = (r, 1)$ and $xy = (1, 1)$. Since $r \neq 1$, the group is nonabelian. This proves existence rather than merely permitting it.

2. All Nontrivial Actions Give One Isomorphism Class

A cyclic group has a unique subgroup of each order dividing its order. Therefore any two order-p elements r,r' of F_q multiplicative generate the same subgroup, and $r' = r^k$ for some k with $1 \leq k < p$.

In the group constructed with parameter r, replace y by y^k. It still generates P because $\gcd(k, p) = 1$, and its action on x is exponent $r^k = r$‘. Consequently the map from the r'-group sending (i,j) to (i, kj) in the r-group respects the multiplication rule and is bijective. All nontrivial actions are therefore isomorphic.

E. N5 — Exhaustion, Distinctness, and Branch Closure

N3 shows every group is one of the action constructions. If p does not divide $q - 1$, only $r = 1$ is possible. If p divides $q - 1$, the trivial and nontrivial constructions both exist, and N4C identifies all nontrivial ones. The two classes are distinct because one is abelian and the other is not. The classification follows. □

□

V. WORKED ORCHESTRATION

- **Order**

  15

  Divisibility test

  3 does not divide 4

  Isomorphism classes

  C15 only

- **Order**

  6

Divisibility test

2 divides 2

Isomorphism classes

C6 and the $r = -1$ semidirect product, isomorphic to S3

- **Order**

  21

  Divisibility test

  3 divides 6

  Isomorphism classes

  C21 and one nonabelian group

For order 21 choose $r = 2$ modulo 7, since $2^3 = 8 \equiv 1$ and $2 \neq 1$. Its relation is $yxy^{(-1)} = x^2$. Choosing r'= 4 instead gives no new class: replace y by y^2. A concrete multiplication is

$$(1, 1)(3, 2) = (1 + 2 \cdot 3, 3) = (0, 0)$$

in residues modulo 7 and 3. These elements are inverses, consistent with the general inverse formula.

A. Boundary and hypothesis audit

The assumption $p < q$ identifies which Sylow subgroup is forced normal. Equal primes lead to order p^2 and a different classification. A nontrivial action does not eliminate the trivial one. Distinct action parameters are not automatically distinct isomorphism classes.

Export normality of Q, the action normal form, semidirect construction, and the full classification. The resulting module demonstrates structural branching, a concrete construction, and equivalence between apparently different solutions.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

A failed branch observes that Sylow allows $n_p = q$ when p divides $q - 1$ and declares the classification finished.

- **Node:** N5

- **State:** STALLED
- **Failing component:** Execution
- **Missing obligations:** ['construct nonabelian group', 'prove all nontrivial actions equivalent', 'retain cyclic class']
- **Repair target:** N4
- **Blocked export:** exactly two isomorphism classes

Knowledge retrieves the automorphism-group structure. Validation distinguishes an allowed count from an actual group. Strategy moves to action parameters. Execution supplies the explicit pair group and generator-change isomorphism. Contract then combines exhaustion, construction, uniqueness, and distinctness. This closes the precise gap left by the shorter classification sketch.

B. Hypothesis boundary

The assumption $p < q$ identifies which Sylow subgroup is forced normal. Equal primes lead to order p^2 and a different classification. A nontrivial action does not eliminate the trivial one. Distinct action parameters are not automatically distinct isomorphism classes.

Export normality of Q, the action normal form, semidirect construction, and the full classification. The resulting module demonstrates structural branching, a concrete construction, and equivalence between apparently different solutions.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. University of Minnesota group theory notes: groups of order pq.